



4.9 Further hypothesis tests: the t-test (HL)

Statistics and probability · IB Diploma · Concept guide and original exam-style practice

Understand the concept

A t-test is used to test claims about a population mean when the population standard deviation is unknown and must be estimated from the sample, particularly with small samples. A one-sample t-test compares a sample mean to a hypothesised value; a two-sample t-test compares the means of two independent samples. The test statistic follows a t-distribution, and using technology to obtain a p-value, H_0 is rejected if the p-value is less than the chosen significance level.

Worked example: follow the reasoning

A nutritionist claims the mean sugar content of a cereal is 12 g per serving. A sample of 10 servings gives a mean of 12.8 g with sample standard deviation 1.5 g, and a t-test (two-tailed) returns a p-value of 0.043. Test at the 5% level whether the mean differs from 12 g.
Solution: $H_0: \mu=12$; $H_1: \mu \neq 12$. Since $p=0.043 < 0.05$, reject H_0 : there is significant evidence at the 5% level that the mean sugar content differs from 12 g.

Quick check · multiple choice

A t-test is used instead of a z-test when:

- A The population standard deviation is known and the sample is large
- B The population standard deviation is unknown and estimated from the sample
- C The data is not numerical
- D The sample size is very large (>1000)

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A gym claims a new training program increases the average bench press by more than 5 kg after 8 weeks. A random sample of 12 members shows a mean increase of 6.2 kg with sample standard deviation 2.1 kg. (a) State the null and alternative hypotheses for an appropriate one-sample t-test. (b) Given a t-test (via technology) returns a p-value of 0.032, state the conclusion at the 5% significance level. (c) State one assumption required for this t-test to be valid. [6]

Final answer: _____



Answers and explanation

Question 1 [6]

(a) $H_0: \mu=5$; $H_1: \mu>5$ (a one-tailed test). (b) Since $p=0.032<0.05$, reject H_0 : there is significant evidence at the 5% level that the mean increase exceeds 5 kg, supporting the gym's claim. (c) The underlying population of increases is assumed to be approximately normally distributed (important given the small sample size), with the sample being random and independent.

Quick check answer

B. A t-test accounts for the extra uncertainty of estimating σ from the sample, especially important for small samples.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.